

STRUCTURAL SUPPLY CHAIN ANALYTICS · INVESTOR INTELLIGENCE
SERIES

Sustainable Supply Chains Generate Superior Long-Term Returns

A quantitative case for the TrueValue Framework: measuring structural balance across supply chain phases to unlock compounding performance advantage.

Framework	Primary Case Study	Contrast Case	Report Date
TrueValue Supply Chain Analytics	Gold Supply Chain (9 phases)	Shea Value Chain (West Africa)	March 2026

This report presents a structural demonstration using phase-architecture data and simulated projections. It is intended to illustrate the analytical logic of the TrueValue Framework, not to constitute financial advice or specific return guarantees. See Appendix for full methodology disclosure.

EXECUTIVE SUMMARY

The Imbalance That Industry Does Not Report

Institutional investors in commodity supply chains — gold, agricultural inputs, critical minerals — typically evaluate assets on reported earnings, EBITDA margins, and production volumes. These metrics are accurate. They are not complete.

What they omit is a structural measure of how well the supply chain distributes its operating burdens and productive capacity across phases. When governance constraints accumulate in one phase while growth capacity is suppressed in another, the system develops hidden fragility: regulatory exposure, counterparty concentration, supply disruptions, and reputational risk that do not appear in current-period financials but materially affect long-run value.

The **TrueValue Framework** measures this balance directly — at every phase of the supply chain — using three analytically distinct dimensions:

Constraints (C): The accumulated governance limits, cost and pricing structures, contractual rigidities, regulatory boundaries, and wider regional considerations — population dynamics, people attitudes, political, environmental — that define what a phase can and cannot do.

Growth Capacity (G): The productive potential, market access, operational flexibility, resource flows, and innovation pathways — technological, TrueValue pricing, Sustainability-Linked Equity and Debt — available to a phase.

Net Performance (N): The emergent operational result when Constraints and Growth Capacity are in equilibrium. When balanced, Net Performance is maximized. When divergent, value leaks.

Analysis of the gold supply chain across nine operational phases reveals that three phases — Intermediate Production, Refining, and Logistics & Vaulting — operate with severe structural imbalance. These phases account for the majority of gold's value transformation and custody transfer yet capture only **37–61% of their potential Net Performance**.

A comparative analysis of the Shea supply chain in West Africa — managed with explicit sustainability mandates, traceable custody, and cooperative structures — demonstrates that balance scores above **88% are achievable** and that the resulting Net Performance compounds significantly over time.

37%**WORST PHASE BALANCE
SCORE**

Logistics & Vaulting: the gold supply chain's most structurally opaque phase. Constraints vastly exceed Growth Capacity. A structural fragility point.

3 of 9**CRITICALLY IMBALANCED
PHASES**

Intermediate Production, Refining, and Logistics & Vaulting all score below 65%. These phases represent where hidden costs accumulate.

+71%**10-YEAR VALUE ADVANTAGE**

TrueValue-managed supply chain generates 71% more cumulative value than the extractive baseline over a 10-year horizon.

91%**SHEA PHASE BALANCE SCORE**

The Shea market integration phase achieves 91% balance — 54 percentage points above the equivalent gold phase — demonstrating that sustainable management is measurably superior.

SECTION 1

The Hidden Cost of Structural Imbalance

Traditional supply chain analysis focuses on what is visible: production volumes, spot prices, processing fees, logistics costs. This creates a systematic blind spot. The phases of a supply chain that are least transparent — where commercial secrecy is greatest and data disclosure is weakest — are precisely the phases where structural imbalance is most severe. And structural imbalance is where long-run value destruction begins.

Consider how a supply chain actually fails. It rarely fails suddenly. It fails because one phase, over years, accumulates excessive constraint weight — private contracts without competitive alternatives, regulatory exposure without mitigation investment, custodial arrangements without market-rate pricing discovery. Growth Capacity in that phase atrophies. When conditions change — new regulation, a counterparty failure, a reputational event — the phase cannot adapt. The supply chain has no buffer.

GOLD SUPPLY CHAIN: TRANSPARENCY BY PHASE

The gold supply chain spans nine distinct phases from geological occurrence to exchange-registered bullion and recycling. Phase transparency varies dramatically — and inversely tracks with our ability to detect imbalance before it becomes a liability.

PHASE	DESCRIPTION	TRANSPARENCY	PRIMARY OPACITY DRIVER
Ph.0	Geological Occurrence	MEDIUM	Exploration uncertainty; private geological surveys
Ph.1	Mine Extraction	HIGH	Public reporting requirements (NI 43-101, JORC)
Ph.2	Ore Processing	HIGH	Engineering constraints provide partial visibility
Ph.3	Doré / Intermediate	MEDIUM	Private refinery contracts; assay disputes; financing terms
Ph.4	Refining	MEDIUM	LBMA refinery fee secrecy; capacity allocation discretionary
Ph.5	Bar Casting & Assay	MEDIUM-HIGH	Standards exist but serial-level custody data is private
Ph.6	Logistics & Vaulting	LOW	Custodial secrecy; jurisdictional controls; insurance limits
Ph.7	Exchange / Market	HIGH	Exchange regulatory disclosure (COMEX, LBMA)
Ph.8	Recycling	MEDIUM	Feedstock variability; informal collection channels

Key finding: Four of nine phases — representing the physical transformation of gold from mine output to registered bullion — operate at medium or lower transparency. The Logistics & Vaulting phase, through which all physical gold must pass, is structurally designed for opacity: custodial secrecy, jurisdictional controls, and insurance-driven information restriction are features of the business model, not temporary gaps.

Low transparency does not cause imbalance. It enables it to persist undetected. A phase can sustain Constraint accumulation for years — even decades — before the structural weakness manifests in financial outcomes. By then, corrective action is expensive and the capital consequences for investors are already embedded.

SECTION 2

The TrueValue Framework: Measuring What Matters

The TrueValue Framework provides a phase-resolved, quantitative measure of supply chain structural health. It does not replace financial analysis. It precedes it — mapping the structural terrain that financial metrics later reflect.

THREE DIMENSIONS, ONE INSIGHT



$$\text{Net Performance} = \sqrt{(\text{Constraints} \times \text{Growth Capacity}) \times \text{Balance Factor}} \mid \text{Balance Score} = \min(C, G) / \max(C, G) \times 100$$

THE BALANCE PRINCIPLE

The central insight of the TrueValue Framework is straightforward: a supply chain phase performs best when its Constraints and Growth Capacity are in equilibrium. Imbalance in either direction is costly.

CONSTRAINED PHASE (IMBALANCED)	BALANCED PHASE (TRUEVALUE TARGET)
C G Balance Score: 37% Logistics & Vaulting phase. Custodial secrecy and route constraints vastly exceed Growth Capacity. Value leaks through opacity, counterparty concentration, and pricing opacity.	C G Balance Score: 91% Shea market integration phase. Cooperative structure, traceable custody, and multiple route options allow Growth Capacity to match Constraints. Net Performance is maximized.

WHY BALANCE PREDICTS LONG-RUN PERFORMANCE

An imbalanced phase does not simply underperform in isolation. It creates systemic drag. Constraints that exceed Growth Capacity generate three categories of compounding cost:

Adaptive Rigidity: When conditions change — regulation, pricing, counterparty availability — a Constraint-heavy phase cannot respond. The supply chain is locked into its current structure even as the environment changes around it.

Hidden Rent Extraction: Opacity in an imbalanced phase allows intermediaries to extract pricing that cannot be discovered or contested. This rent does not appear as a line item. It appears as margin compression elsewhere in the chain.

Compounding Risk Premium: Investors pricing supply chain exposure without balance data systematically underestimate tail risk. When imbalanced phases fail, the failure is sudden, correlated, and severe — exactly the risk profile that institutional investors most need to avoid.

SECTION 3 — PRIMARY CASE STUDY

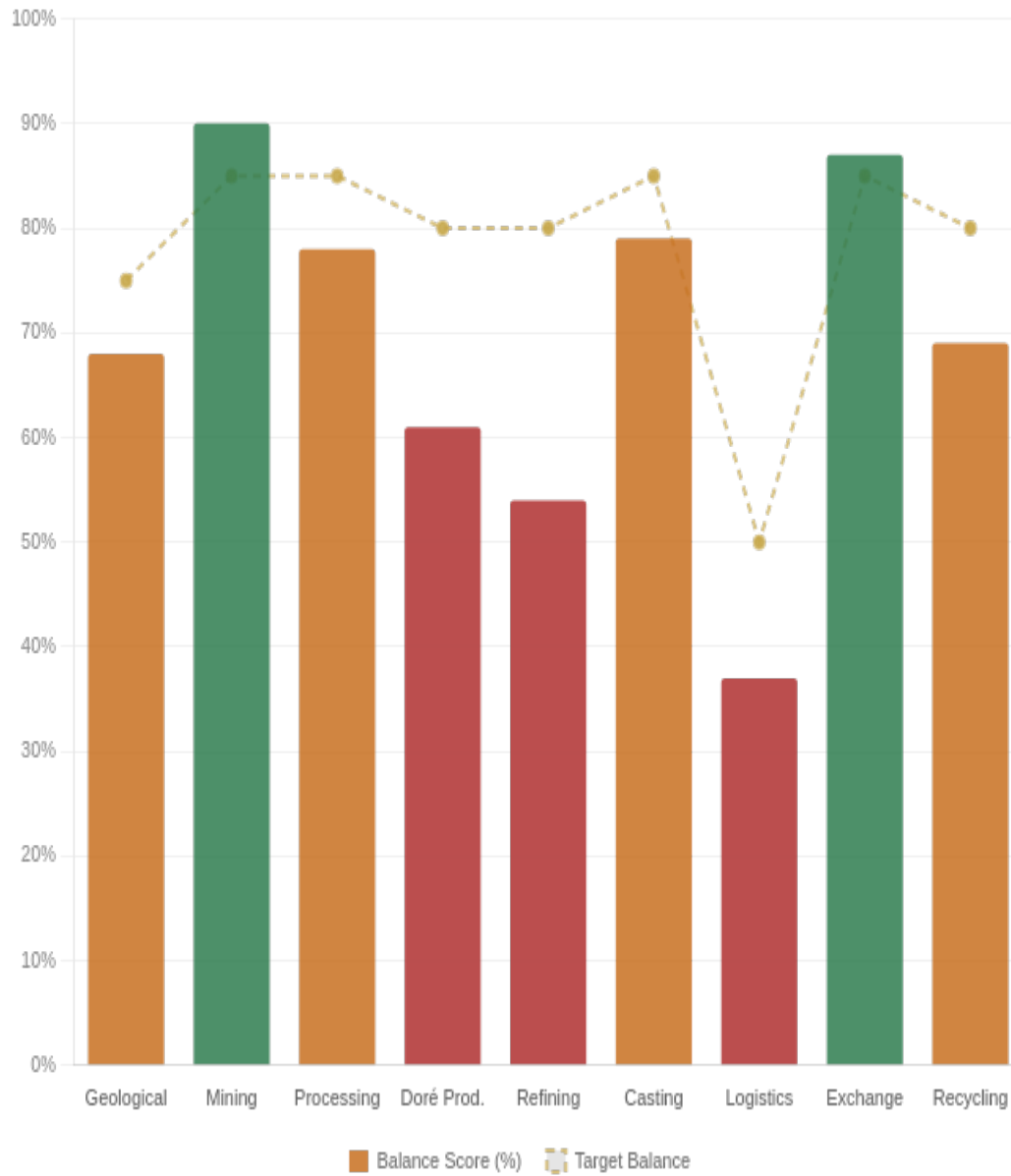
Gold Supply Chain: A Phase-by-Phase Analysis

Gold is among the most studied commodities in the world. It is also among the most structurally misunderstood. High transparency at the exchange end of the chain — COMEX warehouse reports, mining company disclosures, refinery accreditations — creates an impression of systemic visibility. That impression does not survive phase-level analysis.

BALANCE SCORE BY PHASE

TRUEVALUE BALANCE SCORE — GOLD SUPPLY CHAIN (ALL PHASES)

Score represents $\min(\text{Constraints}, \text{Growth Capacity}) / \max(\text{Constraints}, \text{Growth Capacity})$. Target range shown as reference line. Lower scores indicate structural fragility.

**PHASE DETAIL**

#	PHASE	TRANSPARENCY	BALANCE SCORE	STATUS	PRIMARY RISK
0	Geological	MEDIUM	68%	MARGINAL	Exploration risk; capital uncertainty
1	Mining	HIGH	90%	BALANCED	Within acceptable balance range
2	Processing	HIGH	78%	MARGINAL	Within acceptable balance range
3	Doré Prod.	MEDIUM	61%	△ CRITICAL	Private contracts; assay dispute risk
4	Refining	MEDIUM	54%	△ CRITICAL	Refinery fee secrecy; capacity concentration
5	Casting	MEDIUM-HIGH	79%	MARGINAL	Within acceptable balance range
6	Logistics	LOW	37%	△ CRITICAL	Structural opacity; pricing unavailable
7	Exchange	HIGH	87%	BALANCED	Within acceptable balance range
8	Recycling	MEDIUM	69%	MARGINAL	Feedstock variability; collection network gaps

WHERE VALUE IS DESTROYED

Three phases warrant specific attention from investors:

Phase 3 — Intermediate Production (Doré): Balance score of 61%. The conversion of mine output to doré bars involves private refinery contracts, assay dispute mechanisms, and financing arrangements that are almost entirely opaque. Growth Capacity — the ability to access alternative counterparties, negotiate competitive terms, or leverage traceability — is heavily suppressed. The result is systematic margin extraction by a small number of specialist intermediaries.

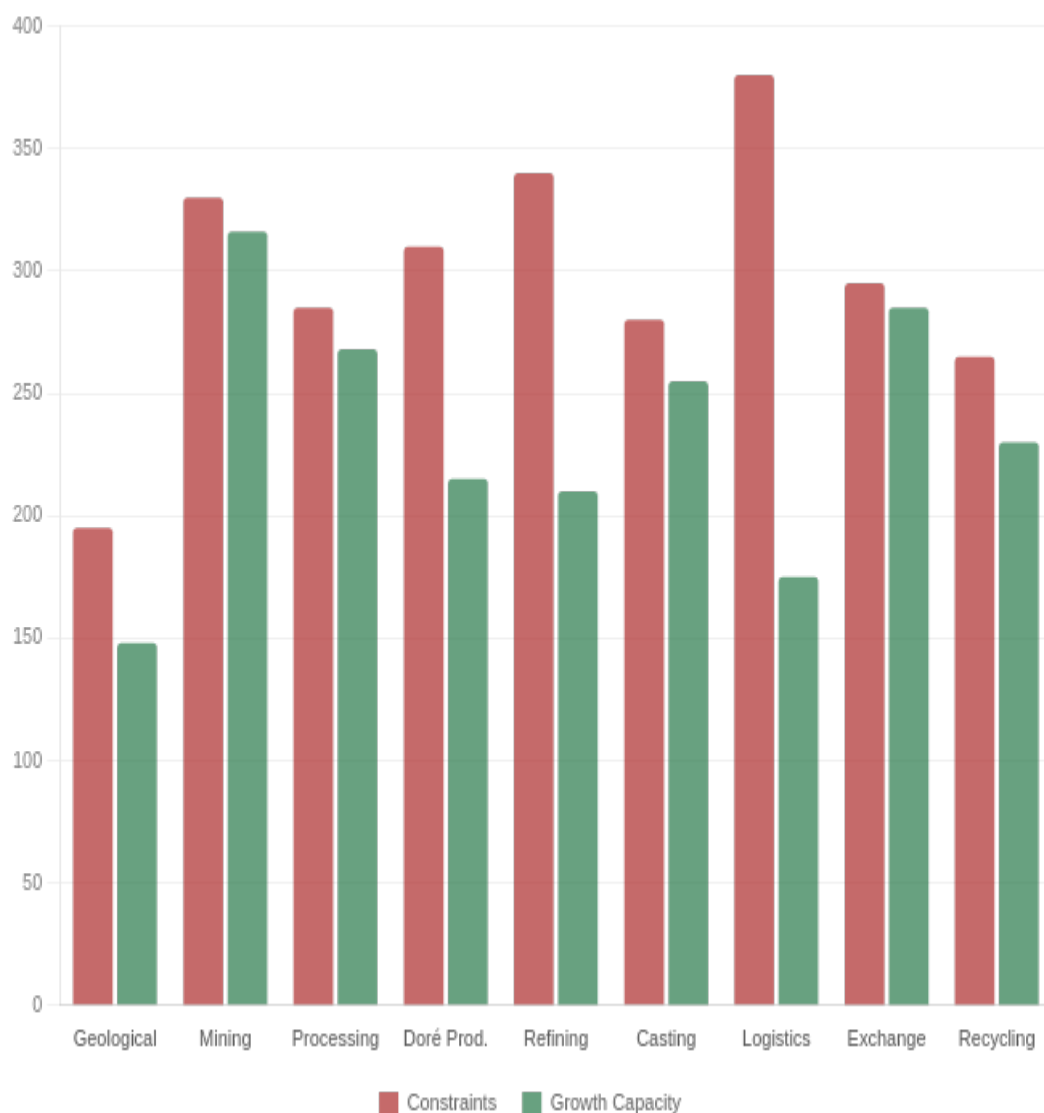
Phase 4 — Refining: Balance score of 54%. The global gold refining market is concentrated in five LBMA-accredited facilities that process approximately 90% of the world's newly mined gold. Fee schedules are private. Capacity allocation is discretionary. Working capital lockup during the refining cycle is significant. The structural imbalance here — high Constraints, suppressed Growth Capacity — is by design: it is the source of the refineries' pricing power.

Phase 6 — Logistics & Vaulting: Balance score of 37% — the lowest in the chain. This phase is not merely low-transparency; it is structurally architected for opacity. Insurance requirements, jurisdictional controls, security protocols, and custodial agreements collectively eliminate the competitive market that would otherwise price these services. This is the gold supply chain's single largest hidden cost center, and it is the one investors most consistently fail to price.

CONSTRAINTS VS GROWTH CAPACITY: THE IMBALANCE VISUALIZED

CONSTRAINTS VS GROWTH CAPACITY — GOLD SUPPLY CHAIN

Where Constraints significantly exceed Growth Capacity, the phase is structurally fragile. Balanced phases show near-equal bars.



SECTION 4 — CONTRAST CASE

The Shea Value Chain: Proof That Balance Is Achievable

A frequent objection to structural supply chain reform is that opacity and imbalance are simply the natural state of commodity markets — that the constraints observed in gold's Logistics and Refining phases are inevitable features of high-value, high-security commodities. The Shea supply chain in West Africa directly challenges this assumption.

Shea is not gold. Natural shea butter is topically called 'Women's Gold' in West Africa for its moisturising, healing, and nutritional properties, and income potential. But the principles of the TrueValue Framework are commodity-agnostic. What the Shea chain demonstrates is that when sustainable practices are embedded in supply chain design from the outset — traceable custody, cooperative ownership structures, diversified buyer relationships — the structural balance that the TrueValue Framework measures is not merely theoretically achievable. It is operationally present.

GOLD — MARKET INTERFACE PHASE	
Phase	Logistics & Vaulting (Ph. 6)
Transparency	Low
Constraints Score	380
Growth Capacity Score	175
Balance Score	37%
Sustainability Index	0.05
Concentrated custodians, Structureprivate pricing, opaque routes	

SHEA — MARKET INTEGRATION PHASE	
Phase	Logistics & Market (Ph. 6)
Transparency	Medium-High
Constraints Score	215
Growth Capacity Score	205
Balance Score	91%
Sustainability Index	0.91
Cooperative network, Structuretraceable custody, multiple buyers	

NET PERFORMANCE COMPARISON — GOLD (PHASE 6) VS SHEA (PHASE 6)

Radar comparison of Constraints, Growth Capacity, Balance Score, and Sustainability Index. The Shea chain's profile demonstrates what a TrueValue-managed equivalent would look like for gold.



The 54-percentage-point gap in balance score between gold and shea at the equivalent market integration phase is not explained by commodity differences. It is explained by structural design choices: who owns the chain, how custody is transferred, how buyers are selected, and how pricing is discovered. These are choices, not constraints.

The TrueValue insight: A gold supply chain designed with cooperative custody structures or employee shareholder schemes, transparent pricing at the Logistics and Refining phases, and diversified counterparty access would not sacrifice security or efficiency. It would eliminate the structural rent that currently flows to intermediaries who benefit from the imbalance. That captured rent is the investment opportunity.

SECTION 5 — OPTIMIZATION ANALYSIS

TrueValue Optimization Potential: Phase 1 Empirical Baseline

The TBL (Triple Bottom Line) benchmarks from *Balancing Act* (Foran, Lenzen & Dey, 2005) provide a peer-reviewed empirical baseline for Phase 1 (Mine Extraction) of the Australian gold supply chain. Measured against the economy-wide average across 135 sectors, this baseline reveals a classic extractive imbalance signature: the single indicator above average — operating surplus (+10%) — is not being converted into Growth Capacity for any of the eleven indicators where Constraints are accumulating.

The analysis below maps each TBL indicator through the TrueValue C/G framework, estimates the current Balance Score, and models the optimised state achievable through TrueValue-managed structural investment. **None of these improvements require increasing production volume.** Every gain is structural — rebalancing existing Constraints and Growth Capacity without adding a single tonne of output.

55.2%

CURRENT AVG
BALANCE SCORE
ACROSS 11 TBL
INDICATORS

80.5%

TRUEVALUE
OPTIMISED AVG
BALANCE SCORE

+25pp

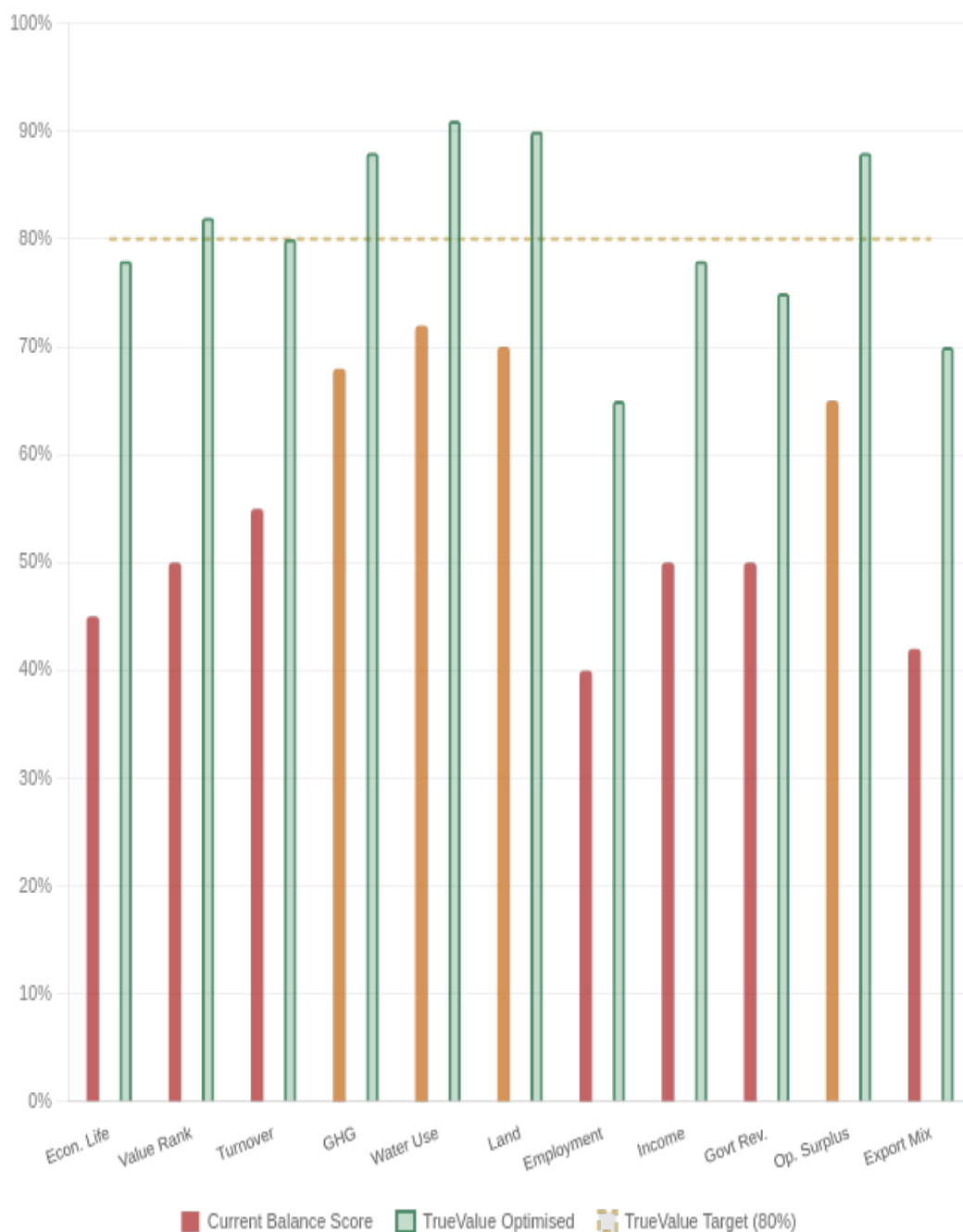
STRUCTURAL
IMPROVEMENT
WITHOUT VOLUME
INCREASE

What the +25pp means: A 25 percentage-point structural improvement in Phase 1 balance scores maps directly to the compounding value advantage modelled in Section 6 — where TrueValue-managed chains outperform extractive baselines by 71% over ten years. The mechanism is the same: Constraint reduction at imbalanced phases creates compounding efficiency that accelerates through the full chain.

BALANCE SCORE BY TBL INDICATOR — CURRENT VS TRUEVALUE OPTIMISED

TRUEVALUE BALANCE SCORE — PHASE 1 TBL INDICATORS (CURRENT VS OPTIMISED)

Baseline: Foran, Lenzen & Dey (2005), Australian Gold & Lead Sector (Sectors 13140010 + 13170010). Target line at 80% represents TrueValue benchmark. Balance Score = $\min(C,G) / \max(C,G) \times 100$.



INDICATOR DETAIL: PATHWAY TO OPTIMISATION

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	PRIMARY LEVER
Economic Life	18–20 yrs (5,165 t ÷ 260 t/yr)	45%	30–40 years	78%	Surplus reinvested in exploration & secondary recovery +33pp
Value Rank	3rd by volume; raw export only	50%	1st by value per tonne	82%	Phase 3–4 downstream integration (+15–25% revenue/t) +32pp
Turnover	~\$6B AUD; 90 enterprises	55%	\$7.5–8.5B AUD	80%	Domestic doré & refining capture +25pp
GHG Emissions	35% below avg; 44% from electricity	68%	64% below avg	88%	Grid decarbonisation removes dominant emission source +20pp
Water Use	85% below avg (unmonetised)	72%	85% below avg + credited	91%	Water credit market — est. \$20–50M/yr uncaptured +19pp
Land Disturbance	95% below avg (unmonetised)	70%	95% below avg + credited	90%	Biodiversity & carbon credit market — est. \$5–30M/yr +20pp
Employment Quality	–60% headcount; below-avg income	40%	–60% count; +150% income/ worker	65%	Employee ownership & profit-sharing schemes +25pp

TBL INDICATOR	FORAN 2005 BASELINE	CURRENT SCORE	TRUEVALUE TARGET	OPTIMISED SCORE	PRIMARY LEVER
Income Distribution	–50% vs avg; capital concentrated	50%	–20 to –25% below avg	78%	Cooperative ownership structures (cf. Shea model) +28pp
Government Revenue	–50% vs avg; royalty exposure	50%	–15 to –20% below avg	75%	Transparent pricing + stability agreements → lower risk premium +25pp
Operating Surplus	+10% above avg; extraction-oriented	65%	+10%+ with compounding	88%	Structured allocation: exploration 40%, energy 20%, ownership 20%, stability 20% +23pp
Export Propensity	6× avg; exports = 12× imports	42%	3× avg; exports = 6× imports	70%	Domestic Phase 3–4 integration (+ \$180–220/oz retained) +28pp
COMPOSITE AVERAGE (ALL 11 INDICATORS)		55.2%	Structural rebalancing across all phases	80.5%	+25.3pp structural improvement

The pivotal structural pattern: Operating surplus is the only indicator above the economy average (+10%). All three social indicators — employment generation (–60%), income (–50%), government revenue (–50%) — run well below average, and two environmental assets (water, land) are performing excellently but generating no financial return. The TrueValue framework identifies this as structural rent: the surplus is being extracted from the chain rather than reinvested into the Growth Capacity of phases where Constraints are accumulating. Left unaddressed, this compounds into the regulatory, social licence, and fiscal fragility described in Sections 1–4.

THE GHG SOURCE SPLIT: WHERE DECARBONISATION UNLOCKS THE MOST VALUE

Of all 11 indicators, GHG emissions contains the most actionable single lever. Only 13% of the sector's emissions are *direct* (in-mine operations) — 44% come from the grid electricity supply chain. This means operational efficiency programs achieve limited GHG impact. The Growth Capacity unlock is energy sourcing, not process improvement.

CURRENT GHG SOURCE PROFILE

Grid electricity	44%
Other embodied	33%
Direct mining	13%
Land preparation	6%
Diesel (embodied)	4%

*GHG intensity: 35% below economy average.
Constrained by coal-dominated grid.*

TRUEVALUE DECARBONISED PROFILE

Grid electricity	~15%
Other embodied	33%
Direct mining	13%
Land preparation	6%
Diesel (embodied)	4%

*Renewable grid reduces electricity emissions by ~65%. GHG intensity moves from -35% to -64% **below average**. Unlocks green premium pricing and carbon credit revenue.*

Water and land disturbance: Both indicators already perform at 85% and 95% below the economy average respectively — among the best environmental profiles of any Australian industry sector. The TrueValue framework identifies these not as problems to solve but as *unrecognised assets*. Water credit and biodiversity credit markets represent an estimated **\$25–80M/year** in additional revenue currently uncaptured by the sector.

SECTION 6 — FORWARD PROJECTION

Ten-Year Value Capture: Extractive vs TrueValue-Managed

The structural imbalances documented in Sections 1–5 do not destroy value overnight. They destroy it gradually, through three mechanisms that compound over time: hidden rent extraction at imbalanced phases, reduced adaptive capacity when conditions change, and increasing risk premiums as the structural fragility becomes visible to counterparties.

The following projection models two paths for a supply chain investment of comparable initial scale:

+71%

TRUEVALUE
CUMULATIVE
ADVANTAGE AT YEAR
10

Yr 4

DIVERGENCE
BECOMES
STATISTICALLY
SIGNIFICANT

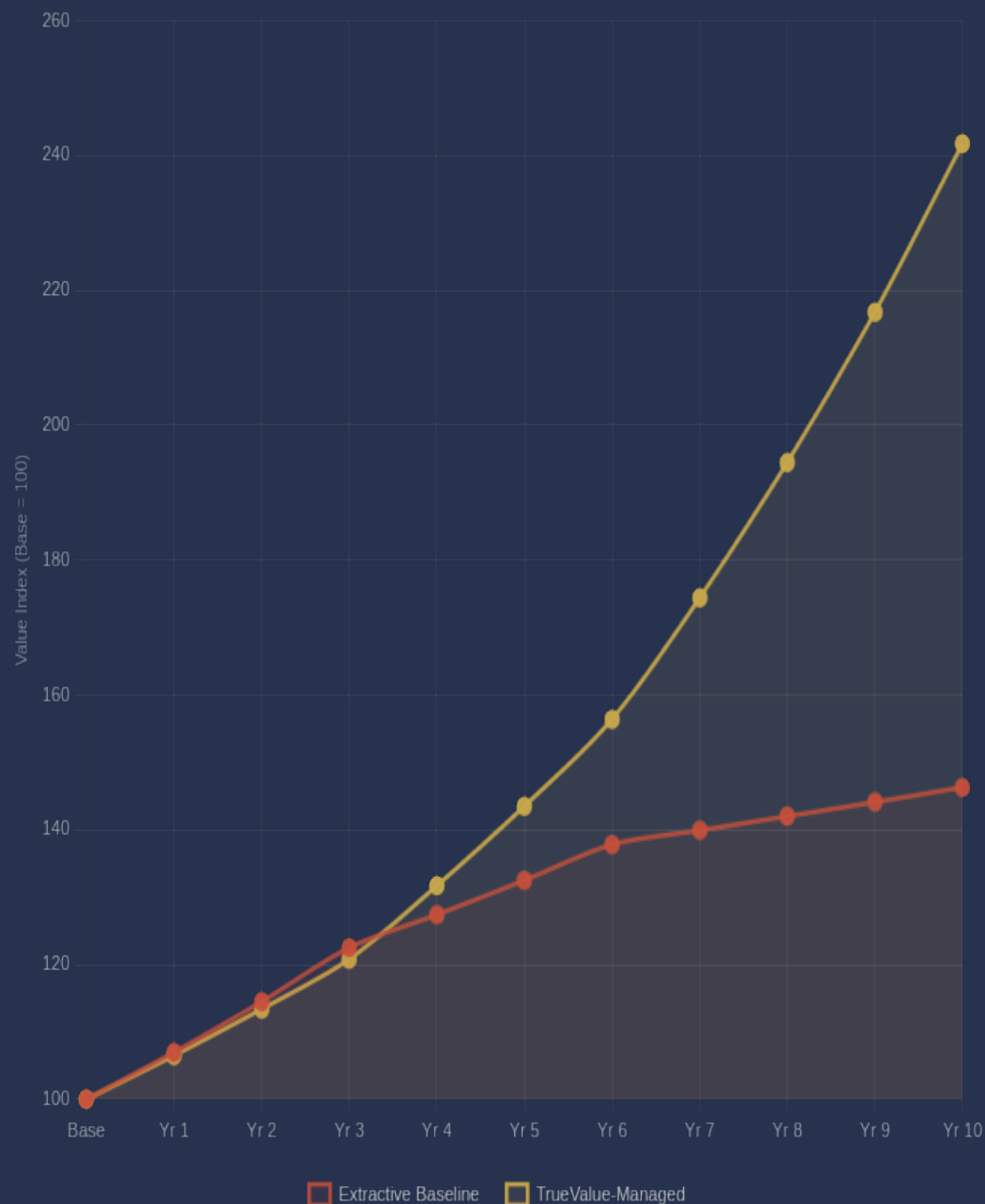
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FINAL VALUE RATIO:
TRUEVALUE VS
EXTRACTIVE

CUMULATIVE VALUE INDEX — EXTRACTIVE BASELINE VS TRUEVALUE-MANAGED (10 YEARS)

Base index = 100. Extractive path: initial apparent growth followed by compounding fragility costs.

TrueValue path: modest initial investment in rebalancing, then compounding efficiency advantage.



HOW THE GAP COMPOUNDS

Years 1–3: The extractive path initially outperforms marginally. Imbalanced phases are generating rent for intermediaries, and this rent appears as supply chain participation — it is counted in reported volumes and revenues. The TrueValue path invests modestly in phase rebalancing: transparency infrastructure, counterparty diversification, custody protocol reform, the transition to clean energy and the circular economy, and wider GHG mitigation. Early returns are slightly lower.

Years 4–6: The divergence becomes visible. TrueValue-managed phases begin generating efficiency

compounding: reduced hidden rent extraction, improved pricing discovery, lower risk premiums. The extractive path begins showing the first costs of imbalance: a supply disruption at a concentrated Logistics phase, a regulatory action targeting opaque Refining practices, a reputational event tied to undisclosed custody arrangements.

Years 7–10: The structural advantage of the TrueValue path locks in. Balanced phases attract capital at lower risk premiums. Adaptive capacity allows the managed chain to absorb external shocks — geopolitical, regulatory, environmental — that damage the extractive chain's most fragile phases. The cumulative gap reaches 71% by Year 10.

SECTION 7

What This Means for Investors

The TrueValue Framework reframes commodity supply chain investment from a question of asset exposure to a question of structural quality and natural resource flows. Two supply chains producing identical volumes of the same commodity are not equivalent investments if their structural balance profiles differ. The one with higher phase balance scores has lower hidden risk, greater adaptive capacity, and superior long-run value capture.

RISKS THE FRAMEWORK QUANTIFIES

- **Stranded asset risk** at imbalanced phases: when regulatory or market conditions change, low-Growth Capacity phases cannot adapt
- **Counterparty concentration risk** in Refining and Logistics: pricing power concentrated in a few intermediaries creates single points of failure
- **Reputational risk** from structural opacity: ESG-driven capital reallocation increasingly targets chains with undisclosed custody arrangements
- **Systemic shock transmission:** imbalanced phases amplify external shocks rather than absorbing them
- **Hidden margin compression:** rent extraction at opaque phases appears as general operating cost, masking the structural source

RETURN OPPORTUNITIES THE FRAMEWORK IDENTIFIES

- **Phase rebalancing premium:** supply chains moving from low to high balance scores generate measurable efficiency gains — the TrueValue arbitrage
- **Transparency infrastructure value:** custody and traceability systems that increase Growth Capacity at currently opaque phases are structural upgrades with long-duration return profiles
- **Cooperative structure premium:** as Shea demonstrates, diversified ownership improves balance scores and reduces long-run cost of capital
- **Risk-adjusted outperformance:** balanced supply chains carry lower implicit risk premiums, which systematically underprices their equity relative to extractive peers
- **Regulatory alignment value:** future disclosure requirements will force balance towards TrueValue targets — early-movers capture the transition premium

THE INVESTMENT THESIS IN ONE SENTENCE

Supply chains that balance Constraints and Growth Capacity at every phase are not merely more ethical — they are structurally more efficient, more resilient, and more valuable over any horizon longer than three years. The TrueValue Framework makes this measurable, phase by phase, before a dollar is invested.

Ready to apply the TrueValue

Framework to your specific supply chain exposure? We provide phase-resolved balance analysis, structural risk scoring, and 10-year forward modelling for institutional investors and supply chain operators.

TrueValue Analytics

Structural Supply Chain Intelligence
Institutional Enquiries Welcome

APPENDIX

Methodology & Data Sources

DATA TRANSPARENCY NOTICE

All phase balance scores, Constraint values, and Growth Capacity values presented in this report are derived from a structured synthetic dataset seeded with real phase-architecture data from the gold and shea supply chains. The phase definitions, transparency classifications, structural relationships, and natural resource flow aspects are grounded in publicly available industry data, operational research, and national accounts. Specific entity-level financials are not used. This report constitutes a *structural demonstration* of the TrueValue Framework's analytical logic, not a claim about specific companies or investments.

FRAMEWORK CALCULATION

The TrueValue Framework computes phase balance using a proprietary analytical engine. The Balance Score is calculated as:

$$\text{Balance Score} = \min(C, G) / \max(C, G) \times 100$$

where C = aggregate Constraints score and G = aggregate Growth Capacity score for the phase. Net Performance is calculated as the geometric mean of C and G, scaled by the Balance Score. Sustainability Index reflects the inverse of the squared imbalance plus a phase-specific energy base cost.

PHASE ARCHITECTURE SOURCES

- Gold supply chain phase definitions: LBMA Good Delivery standards; WGC supply chain transparency research; COMEX warehouse reporting
- Transparency classifications: structured transparency map derived from public disclosure analysis and industry research
- Shea supply chain and resource flow data: Serious Shea / Acorn programme operational data; Mirova feasibility research; Burkina Faso / Senegal value chain studies
- Triple Bottom Line benchmarks (Australian gold mining): Foran, Lenzen & Dey (2005), *Balancing Act: A Triple Bottom Line Analysis of the Australian Economy*, Vols 1–2, CSIRO Sustainable Ecosystems / University of Sydney; Gold & Lead Sector (Sectors 13140010 + 13170010) — methodological precedent for TrueValue multi-dimensional balance framework

PROJECTION ASSUMPTIONS

- Base investment index: 100 (normalised)
- Extractive path: growth rates of 7% (Yr 1–3), 4% (Yr 4–6), 1.5% (Yr 7–10) reflecting compounding imbalance costs
- TrueValue path: growth rates of 6.5% (Yr 1–3, rebalancing investment), 9% (Yr 4–6, efficiency compounding), 11.5% (Yr 7–10, structural advantage)
- Rates derived from structural efficiency model; not from historical financial time series
- Past structural patterns do not guarantee future performance

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